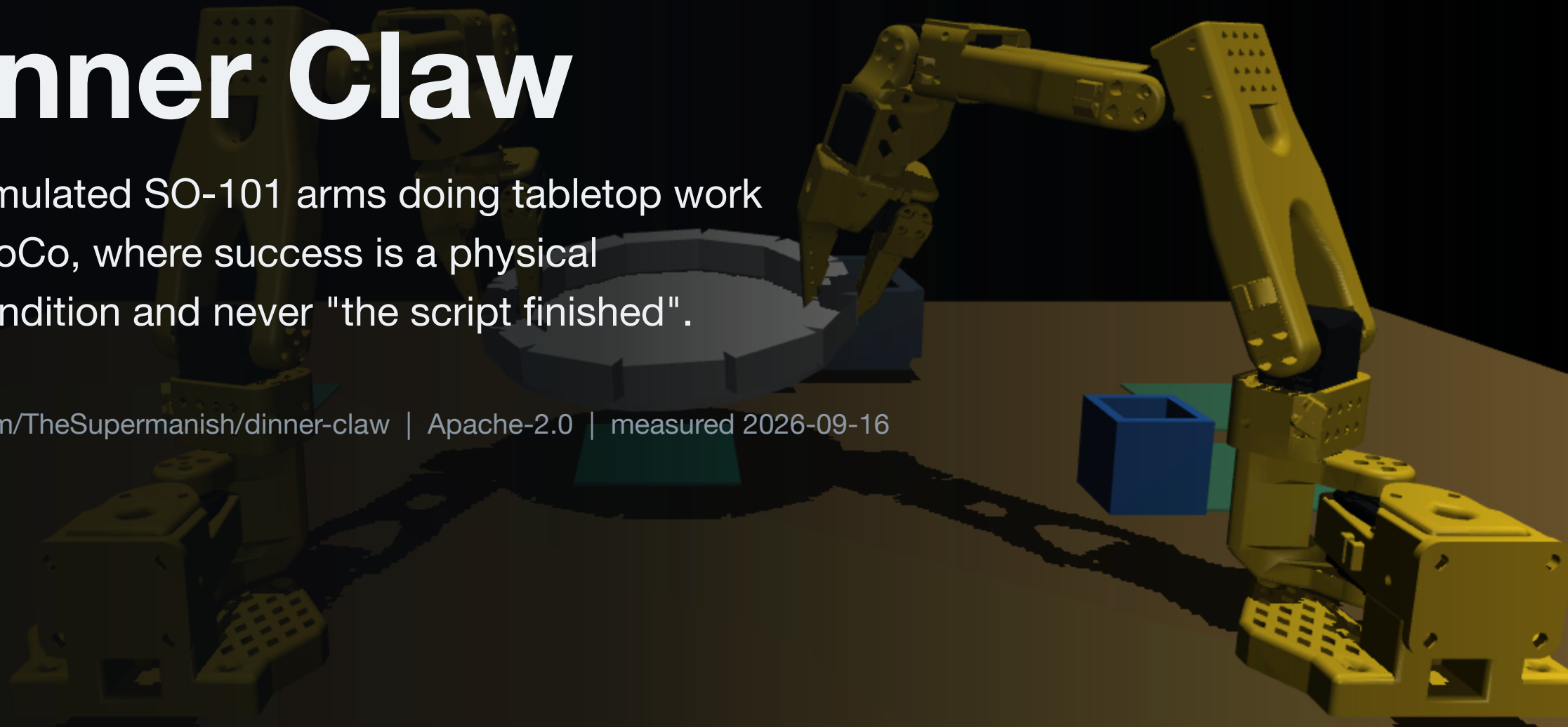


INTEL PHYSICAL AI ONLINE CHALLENGE

Dinner Claw

Two simulated SO-101 arms doing tabletop work in MuJoCo, where success is a physical postcondition and never "the script finished".

github.com/TheSupermanish/dinner-claw | Apache-2.0 | measured 2026-09-16



THE ARGUMENT FOR TWO ARMS

One arm cannot lift a plate. It levers it.

A single gripper pinches the rim **94 mm** from the plate's centre of mass, so it applies almost pure torque. The plate tips up on its far rim, the centre height rises, and the plate never leaves the table.

That single-arm teacher is **kept in the tree at 0/10** and runs in the demo on its own hotkey. It is the evidence for the whole design, not a result to hide.

0 / 10

one arm, seeds 40-49

9 / 10

two arms, held-out seeds 50-59

6 of 10

single-arm seeds where the plate never left the table at all

Five physical skills, each with a negative control

SKILL	ARMS	SEEDS	RESULT	OPEN-GRIPPER CONTROL
Drawer open	left	30-39 held-out	10/10	0/3
Drawer then fork, one episode, no reset	left	40-49 held-out	10/10	0/3
Plate lift, two arms	both	50-59 held-out	9/10	0/4
Cup place, scripted teacher	right	200-209	10/10	0/3
Cup place, camera-guided	right	500-549	50/50	
Plate lift, ONE arm	left	40-49	0/10	kept deliberately

Say **9/10** for the two-arm lift, not 10/10. Across every configuration measured it lifts and carries level on 20 of 20 seeds and fails the RELEASE on 2, and which seed fails moves when unrelated parts of the scene change.

WHAT "SUCCESS" MEANS HERE

A gate the robot has to pass with physics

- **Force-bearing contact** from the named gripper pads, read through `mj_contactForce`, not proximity.
- **Sustained lift** with nothing outside the robot bearing the weight, so a plate still resting on the table cannot score.
- **Stable release** at resting height after the gripper opens.
- **Levelness**, for the two-arm lift: over **15 degrees** at any airborne moment and the episode fails.

No welds, no teleports, no raised motor force limits, and the drawer joint is never driven directly. Workcell setup happens only before an episode and is declared in the result.

Failed episodes are preserved and labelled, not deleted.

4 defects

found by adversarial review of these gates, each a gate that was necessary but not sufficient. All four fixed before this deck.

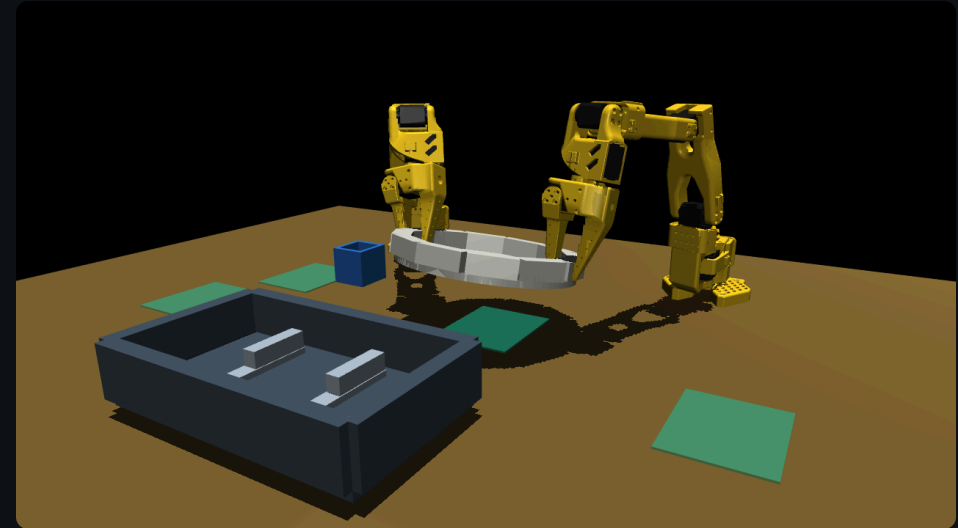
THE COOPERATIVE ACTION

Two arms cancel the torque one arm creates

Both arms close on **opposing rim segments**, 188 mm apart. The two contact wrenches cancel about the plate's centre, so the net action is a lift rather than a pivot.

The plate then travels to a mat inside the two-arm overlap and is released standing.

- Peak lift **57-60 mm**, peak tilt while airborne **12.3-13.8 deg**, across the ten held-out seeds
- Four-pad force-bearing contact **9.1 s**, of which **4.6 s** airborne and level
- Released on the mat within **4-25 mm** of its centre



Real captured frame, seed 50, at the instant the episode's own gate reports the plate airborne and level.

NEGATIVE CONTROLS

Every skill has a run that must score zero

The same teacher runs with the grippers **held open**. If it still scores, the gate was measuring the script, not the robot.

0 successes across all four controls, and each one is rejected at its **first** physical gate rather than drifting to a later failure with the same status.

drawer 0/3

rejected at the handle pull

fork 0/3

rejected at the grasp

plate 0/4

rejected at the four-pad close gate

cup 0/3

rejected at the grasp

The test suite asserts the PHASE each control dies in, so a disabled gate fails the suite rejected at the gate

CHAINING WITHOUT A RESET

The fork comes out of the drawer the robot just opened

One episode, no reset between skills, no weld, no teleport. The drawer stage must first travel **at least 8.5 cm** through a verified bilateral handle pull, and then stay open with the **whole arm off the whole drawer**: an arm leaning on the drawer front fails the gate.

Only then does the fork retrieval run, and it depends on the physical state the first half produced.

10 / 10

held-out seeds 40-49, drawer then fork in one episode

9.9 cm

typical drawer travel, against an 8.5 cm bound

46 s

wall-clock at 1.0x realtime, the longest skill in the demo

Calibrated vision, and a planner that fails closed

Camera-guided cup place: 50/50. The cup is located from top-camera pixels by calibrated colour segmentation, and the skill runs from that estimate. The policy is given no object poses.

The planner is a deterministic grammar, not a VLM. It previews a skill sequence, derives which arm owns each object from the measured ownership table, and then refuses to execute anything without a measured held-out success rate.

Ask it to pour and it previews `pick(right, bottle) -> pour(right, bottle)` and

then declines to run it. Failing closed is the

`src/tabletop_vla/planning/task_planner.py`; every ownership row takes reproducible evidence, enforced by a test

feature.

50 / 50

camera-guided cup place, seeds 500-549

Not a VLM

SmoIVLM's local scene descriptions were unreliable and it issues no motor commands, so it drives nothing in this system.

ACT trained locally, exported to OpenVINO

- **LeRobot 0.6.0 ACT**, frozen pretrained ResNet18, RGB plus 12 joint positions, 20-action chunks.
- Normalization from training episodes only; validation and test seeds cannot overlap training.
- Exported to **OpenVINO FP32** and running locally, past numerical parity fixtures.
- The **FP16 candidate failed parity and is rejected**. INT8 is not done.

The demo still runs the proven teacher. A learned policy gets promoted when its held-out

physical rollouts justify it, not when its offline

loss looks good

No Intel result

Everything here ran on Apple silicon. Mac CPU inference numbers are diagnostics and are not challenge results. The qualifying Core Ultra Series 2/3 run has not happened.

The honest half of the scoreboard

- **Pouring.** The bottle now has a graspable 32 mm neck, and a kinematic probe found **17 graspable bottle poses** with viable bottle/mug pairs up to 216 mm apart. There is still no pour skill, and MuJoCo has no liquid in this scene.
- **The spoon handoff.** Gates are wired and demand a measured receiver lift, but no passing run exists.
- **Chaining beyond drawer to fork.** The full dinner-table sequence is not implemented.
- **INT8 quantization** and the **Intel Core Ultra run.**

17 poses

bottle neck grasps found by the probe, position error under 1.6 mm. Feasible is not implemented.

A gate with no passing run behind it is listed here, not on the results slide.

LIVE DEMO

Native MuJoCo viewer, nothing stage-managed

```
uv run mjpgpython -m tabletop_vla.native --  
seed 0
```

- **1** drawer **2** drawer then fork **3** two-arm plate
- **4** one-arm plate, which fails on purpose **5** camera cup
- **0** reset . pause - / = change seed live

Physics keeps stepping after an episode ends, so anything the script was merely holding up falls on camera. That is deliberate.

Seed 0

passes all four skills, verified end to end

1.0x

realtime, no speed-up in the recording

If a skill fails live, the overlay's failure reason is the real output and gets read out as-is.

DINNER CLAW

Two arms, five gated skills, and a scoreboard that includes the zeros.

github.com/TheSupermanish/dinner-claw · Apache-2.0 · MuJoCo
3.13.0, LeRobot ACT, OpenVINO · SO-101 arms

Every figure in this deck reproduces from `outputs/regression-2026-09-16/` on Apple silicon. No Intel hardware result exists yet.